Correcting for lung ventilation in model-based CT

Ryan Andosca, MS[1], Peter Boyle, MS[1], Michael Lauria, PhD[1], Claudia R. Miller, BS[1], Louise Naumann[1], Minji Kim[1], Dylan P. O'Connell[1], Ricky Savjani, MD, PhD[2], Drew Moghanaki, MD, MPH[2] and Daniel A. Low, PhD[1]

(1)Department of Radiation Oncology, University of California, Los Angeles, Los Angeles, CA

(2)University of California, Los Angeles, Los Angeles

AAPM 2025
JULY 27–30 | WASHINGTON, DC
67TH ANNUAL MEETING & EXHIBITION
COMING TOGETHER TO FORGE AHEAD IN MEDICAL PHYSICS

INTRODUCTION

Four-dimensional computed tomography (4DCT) is a standard tool for managing respiratory motion in radiation therapy. However, irregular breathing often leads to image artifacts, which can compromise the accuracy of treatment planning.

To address these limitations, our group developed a model-based approach called 5DCT. This method uses a series of fast-helical, free-breathing CT scans to create a more comprehensive model of lung motion. While 5DCT improves upon the geometric accuracy of the lungs, the current model does not account for changes in lung density.

This work introduces and validates a method to correct for these ventilation-induced density changes within the 5DCT framework. Our goal is to produce more accurate CT images that reflect both the position and density of lung tissue at any point in the breathing cycle, ultimately leading to more precise radiation therapy delivery.

AIM

To develop and validate a Hounsfield Unit (HU) correction method to account for lung ventilation in model-based CT (5DCT).

METHOD

This retrospective study utilized CT datasets from 220 patients, each acquired with a 5DCT imaging protocol consisting of 25 sequential, low-dose, fast-helical scans.

- **Image Registration**
  - A two-stage deformable image registration (deedsBCV followed by pTV) was performed to align all 25 scans to a common reference image, ensuring accurate voxel-wise correspondence.
- **Ventilation Correction Algorithm**
  - For each voxel within the segmented lungs, a linear regression was performed between its Hounsfield Unit (HU) values across all 25 scans and the corresponding normalized respiratory amplitude. The HU values corrected to an amplitude A is:

$$HU_{corrected} = \frac{dHU}{dA}\left(A - A_{ref}\right) + HU_{ref}$$

    where A_{ref} and HU_{ref} are the breathing amplitude and HU of the voxel in the reference image.
  - The slope (dHU/dA) defined the voxel-specific ventilation. Each breathing range was normalized to 0 to 1, corresponding to the 5th and 95th percentile amplitudes, respectively.
  - To improve accuracy, the data was smoothed with a 3x3x3 voxel average, and statistical outliers were iteratively removed before fitting.
  - Voxels with non-physical slopes, often resulting from registration errors or high-contrast gradients, had their slopes replaced by an interpolated value from the nearest valid neighbor.
- **Validation via Fluence Simulation**
  - The method was validated by simulating the transmission of a 6 MV parallel-ray photon beam through the lungs in the anterior-posterior direction.
  - Fluence differences were calculated for all pairs of scans, both with and without the ventilation correction applied.
  - To minimize the impact of potential registration errors near the diaphragm, the analysis was repeated with the inferior portion of the lungs cropped.

RESULTS

Our ventilation model successfully characterized density changes in lung parenchyma.

Ventilation Model Findings

- The mean ventilation-induced HU change in lung parenchyma was -43 ± 16 HU per normalized breath, compared to just -7 ± 5 HU in soft tissue.
- The linear fit was robust, with a low mean RMS residual of 11 ± 2 HU.
- On average, only 1.3% of lung voxels required interpolation, indicating high model stability.

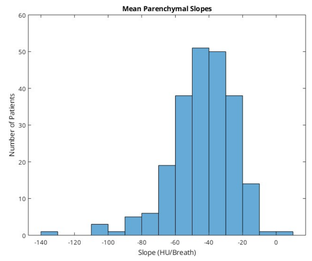


Figure 3 (left): *Distribution of average parenchymal ventilations (slopes). Parenchymal voxels were assumed to be any voxels for which the reference image voxel had an HU value of less than -50 HU*

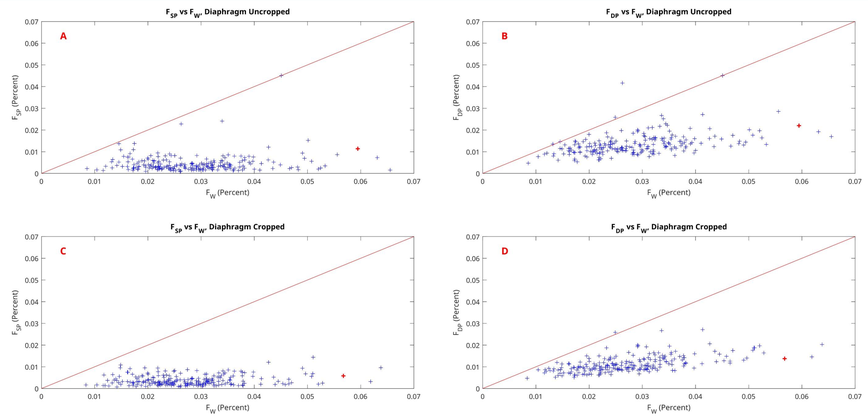


Figure 4 (above): *Shows the plots for the worst-case scan-pair fluence differences, corrected vs. uncorrected, for all patients. A and B show the difference taken at the same patch before and after ventilation correction with the diaphragm region included and excluded respectively; C and D show a comparison between the worst uncorrected patch and the worst corrected patch with the diaphragm region included and excluded respectively. All results that exclude the diaphragm region show marked improvement with ventilation correction, and most results that include the diaphragm region still show significant improvement.*

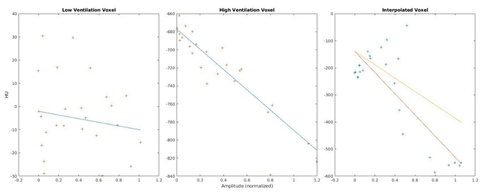


Figure 1 (above): Example *voxel fits: no-ventilation fit (left), highly ventilating fit (middle), and interpolated fit (right).*

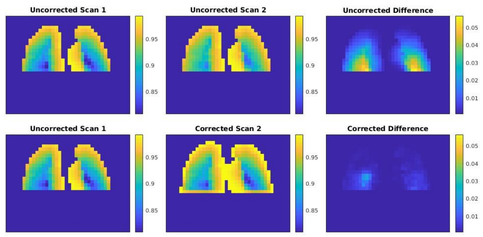


Figure 2 (above): *The first column images show the same uncorrected fluence map for one scan of the patient. The second column shows the uncorrected fluence map (top) and the corrected fluence map (bottom) of another scan. The third column shows the differences between the other two images in the row.*

CONCLUSIONS

- The correction significantly improves the consistency of lung HU values across scans, reducing a potential source of error in radiation therapy planning.
- This work establishes a foundation for more accurate dose calculations for tumors affected by respiratory motion. Future efforts will focus on clinical translation and algorithm refinement.

REFERENCES

[1] Vishnevskiy, V., Gass, T., Szekely, G., Tanner, C. & Goksel, O. Isotropic Total Variation Regularization of Displacements in Parametric Image Registration. *IEEE Trans Med Imaging* **36**, 385–395 (2017).

[2] Heinrich, M. P., Maier, O. & Handels, H. Multi-Modal Multi-Atlas Segmentation Using Discrete Optimisation and Self-Similarities. http://ceur-ws.org.

[3] Heinrich, M. P., Jenkinson, M., Brady, M. & Schnabel, J. A. MRF-based deformable registration and ventilation estimation of lung CT. IEEE Trans Med Imaging 32, 1239–48 (2013).

[4] Thomas, D. H. *et al. Investigating the Minimum Scan Parameters Required to Generate Free-Breathing Motion Artefact-Free Fast-Helical CT*. (2018).

[5] Thomas, D. *et al.* A novel fast helical 4D-CT acquisition technique to generate low-noise sorting artifact-free images at user-selected breathing phases. *Int J Radiat Oncol Biol Phys* **89**, 191–198 (2014).

[6] O'Connell, D. *et al.* Safety-oriented design of in-house software for new techniques: A case study using a model-based 4DCT protocol. *Med Phys* **46**, 1523–1532 (2019).

CONTACT INFORMATION

Email: randosca@mednet.ucla.edu